APPENDIX 1

## Acronyms, Abbreviations, and Glossary

| | |
|---|---|
| AA | Associate Administrator |
| AAA | Active Acquisition Aid |
| AAP | Apollo Applications Program |
| ACS | attitude control system |
| ACN | Ascension |
| ACU | antenna control unit |
| AES | Apollo Extension System |
| AFSC | Air Force Systems Command |
| AFSCN | Air Force Satellite Control Network |
| AGO | Santiago |
| AIS | Apollo Instrumentation Ship |
| ALT | Approach and Landing Test |
| CDR | Critical Design Review |
| CDSCC | Canberra Deep Space Communication Complex |
| CSIR | Council for Scientific and Industrial Research |
| CSIRO | Commonwealth Scientific and Industrial Research Organization |
| CSM | Command and Service Module |
| CSOC | Consolidated Space Operations Contract |
| DAF | Data Acquisition Facility |
| dB | decibels |
| DDMS | Department of Defense Manager for Manned Spaceflight |

| | |
|---|---|
| DF | direction finding |
| DJS | Dzhusaly, Razakhgtan |
| DLR | Germany's Deutsches Zentrum für Luftund Raumfahrt |
| DOD | Department of Defense |
| DOI | Department of the Interior |
| DOS | Department of Supply |
| DRSS | Data Relay Satellite System |
| DSN | Deep Space Network |
| DSS | Deep Space Station |
| EGO | Eccentric Geophysical Observatory |
| EGR | Eglin Gulf Test Range |
| ELVIS | Enhanced Launch Vehicle Imaging System |
| ERS | Earth Resource Satellite |
| ERTS | Earth Resource Technology Satellite |
| ESA | European Space Agency |
| ESD | Air Force Electronic Systems Division |
| ESMC | Air Force Eastern Space and Missile Center |
| EUMETSAT | European Organization for the Exploitation of Meteorological Satellites |
| ETR | Eastern Test Range |
| EVA | extravehicular activity |
| EUT | Eupatona, Ukraine |
| FAA | Federal Aviation Administration |
| FCC | Federal Communications Commission |
| FDR | Final Design Review |
| FM | frequency modulation |
| FY | fiscal year |
| Gbps | gigabits per second |
| GDSCC | Goldstone Deep Space Communication Complex |
| GHz | gigahertz |
| GLTN | Goddard Laser Tracking Network |
| GMT | Greenwich Meridian Time |
| GN | Ground Network |
| GOES | Geostationary Observational Environmental Satellite |
| GRARR | Goddard Range And Range Rate |
| GRGT | Guam Remote Ground Terminal |
| GRO | Gamma Ray Observatory |
| GRTS | GRO Remote Terminal System |
| GSFC | Goddard Space Flight Center |
| GWM | Guam |
| HAW | Hawaii |
| HDTV | high definition television |
| HEAO | High Energy Astronomy Observer |

| | |
|---|---|
| HST | Hubble Space Telescope |
| HTV | H-II Transfer Vehicle |
| IAGS | Inter-American Geodetic Survey |
| IBM | International Business Machines |
| IBEW | International Brotherhood of Electrical Workers |
| ICBM | intercontinental ballistic missile |
| IDEA | International Space Station Downlink Enhancement Architecture |
| IGY | International Geophysical Year |
| IMP | Interplanetary Monitoring Platform |
| INEN | Integrated Near-Earth Network |
| INTA | Spanish National Institute of Aerospace Technology |
| INTELSAT | International Telecommunications Satellite Consortium |
| IOAG | Interagency Operations Advisory Group |
| IPF | Image Processing Facility |
| ISIS | International Satellite for Ionospheric Studies |
| ISO | Infrared Space Observatory |
| ISS | International Space Station |
| IUE | International Ultraviolet Explorer |
| IUS | Inertial Upper Stage |
| JAXA | Japan Aerospace Exploration Agency |
| JDMTA | Jonathan Dickinson Missile Tracking Annex |
| JPL | Jet Propulsion Laboratory |
| JSC | Johnson Space Center |
| JWST | James Webb Space Telescope |
| KHz | kilohertz |
| KLP | Kolpashevo, Russia |
| Kpbs | kilobits per second |
| KPGO | Kokee Park Geophysical Observatory |
| KSAT | Kongsberg Satellite Services |
| KSC | Kennedy Space Center |
| LAGEOS | Laser Geodynamics Satellite |
| laser | light amplification by stimulated emission of radiation |
| LM | Lunar Module |
| LRC | Langley Research Center |
| LRO | Lunar Reconnaissance Orbiter |
| LRV | Lunar Roving Vehicle |
| M&DO | Mission and Data Operations |
| MA | Mercury-Atlas |
| MA | Multiple Access |
| MAD | Madrid |
| maser | microwave amplification by stimulated emission of radiation |
| Mbps | megabits per second |
| MCC | Mission Control Center |

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|---|---|
| MDSCC | Madrid Deep Space Communication Complex |
| MESA | Modular Equipment Stowage Assembly |
| MFS | Manned Flight Support |
| MGS | McMurdo Ground Station |
| MHz | megahertz |
| MILA | Merritt Island Launch Annex |
| MIT | Massachusetts Institute of Technology |
| MOBLAS | Mobile Laser Ranging System |
| MOCR | Mission Operations Control Room |
| MOTS | Minitrack Optical Tracking System |
| MRCS | Mobile Range Control System |
| MSC | Manned Spacecraft Center |
| MSFN | Manned Space Flight Network |
| NACA | National Advisory Committee for Aeronautics |
| NAR | Non-Advocate Review |
| NAS | National Academy of Science |
| NASA | National Aeronautics and Space Administration |
| NASCOM | NASA Communications Network |
| NASDA | National Space Development Agency of Japan |
| NC | Network Controller |
| ND | Networks Directorate |
| NEN | Near-Earth Network |
| NFL | Newfoundland |
| NISN | NASA Integrated Services Network |
| NOAA | National Oceanic and Atmospheric Administration |
| NOCC | Network Operations Control Center |
| NORAD | North American Aerospace Defense Command |
| NRD | Air Force National Range Division |
| NRL | Naval Research Laboratory |
| NSC | Norwegian Space Center |
| NSF | National Science Foundation |
| NTSC | National Television System Committee |
| NTTF | Network Test and Training Facility |
| OAO | Orbiting Astronomical Observatory |
| OCC | Operations Control Center |
| OGO | Orbiting Geophysical Observatory |
| OMS | Orbital Maneuvering System |
| ORR | Orroral |
| OSC | Office of Space Communications |
| OSO | Office of Space Operations |
| OSO | Orbiting Solar Observatory |
| OSTDA | Office of Space Tracking and Data Acquisition |
| OTDA | Office of Tracking and Data Acquisition |

| | |
|---|---|
| PAL | Phase Alternating Line |
| PAM | Pulse Amplitude Modulation |
| PCM | Pulse Code Modulation |
| PCR | Payload Changeout Room |
| PCS | Pacific Command Ship |
| PCU | Power Control Unit |
| PDL | Ponce de Leon Station |
| PDR | Preliminary Design Review |
| PER | Pre-Environmental Review |
| PLSS | Portable Life Support System |
| PMR | Pacific Missile Range |
| PMRF | Pacific Missile Range Facility |
| POCC | Project Operations Control Center |
| POGO | Polar Orbiting Geophysical Observatory |
| PPK | Petropavlovsk-Kamchatskaya, Russia |
| PSAC | President's Science Advisory Committee |
| QUI | Quito |
| RCA | Radio Corporation of America |
| RCS | Reaction Control System |
| RF | radio frequency |
| RFP | Request For Proposal |
| ROS | Rosman |
| RTG | Radioisotope Thermoelectric Generator |
| RTHU | roll to heads-up |
| RTLS | Return to Launch Site abort |
| SA | Single Access |
| SAO | Smithsonian Astrophysical Observatory |
| SATAN | Satellite Automatic Tracking Antenna |
| SCAMA | Switching, Conferencing And Monitoring Arrangement |
| SCAMP | Satellite Command Antenna on Medium Pedestal |
| SDK | Ussuriysk, Russia |
| SDO | Solar Dynamics Observatory |
| SEB | Source Evaluation Board |
| SECAM | Sequential Color with Memory |
| SECO | sustainer engine cutoff |
| SGL | Space to Ground Link |
| SGLT | Space to Ground Link Terminal |
| SGS | SvalSat Ground Station |
| SIRTF | Space Infrared Telescope Facility |
| SITE | Satellite Instructional Television Experiment |
| SN | Space Network |
| SNIP | Space Networks Interoperability Panel |
| SOMO | Space Operations Management Office |

| | |
|---|---|
| SRB | Solid Rocket Booster |
| STADAN | Satellite Tracking And Data Acquisition Network |
| STADIR | Station Director |
| STDN | Spaceflight Tracking and Data Network |
| STG | Space Task Group |
| STGT | Second TDRSS Ground Terminal |
| STS | Space Transportation System |
| STTCS | S-band Tracking, Telemetry & Command System |
| T&DS | Tracking and Data Systems |
| TAGIU | Tracking And Ground Instrumentation Unit |
| TAGS | Text And Graphics System |
| TAL | Transatlantic Abort Landing |
| TAN | Tananarive |
| TAT-1 | trans-Atlantic telephone cable |
| TBL | Tbilisi, Georgia |
| TCS | Thermal Control System |
| TDRS | Tracking and Data Relay Satellite |
| TDRSS | Tracking and Data Relay Satellite System |
| TDSD | Tracking and Data Systems Directorate |
| TELOPS | Telemetry Online Processing System |
| TEI | Trans-Earth Injection |
| TIROS | Television Infrared Observation Satellite |
| TLI | Trans-Lunar Injection |
| TLRS | Transportable Laser Ranging System |
| TM | telemetry |
| TTS | Test and Training Satellite |
| TSSG | Tracking System Study Group |
| TT&C | Tracking, Telemetry and Command |
| TWT | traveling wave tube |
| UHF | ultrahigh frequency |
| ULD | Ulan-Ude, Russia |
| US | United States |
| USAF | United States Air Force |
| USB | Unified S-Band |
| USN | Universal Space Network |
| USNS | United States Navy Ship |
| USSR | Union of the Soviet Socialist Republic |
| VERLORT | Very Long Range Tracking |
| VHF | very high frequency |
| VLBI | Very Long Baseline Interferometry |
| WART | White Sands Complex Alternative Resource Terminal |
| WECO | Western Electric Company |
| WGTC | Working Group on Tracking and Computation |

| | |
|---|---|
| WRE | Weapons Research Establishment |
| WSGT | White Sands Ground Terminal |
| WSMR | White Sands Missile Range |

## Glossary

**acquisition**: The process of first finding, either visually or electronically, a satellite or spacecraft of interest so it can then be tracked.

**antenna feed**: The electronic device at the focal point of an antenna through which electromagnetic wave transmissions are received, amplified, and/or transmitted.

**apogee:** The high point in a trajectory.

**array:** The electronic combining of antennas pointed at the same object so as to increase the received signal strength.

**autotrack:** Automatic tracking of a spacecraft by an antenna (or vise versa) where the position is continuously and automatically computed.

**bandwidth:** The range of frequencies occupied by a radio frequency carrier wave. The more information there is on the carrier (for example, high-definition television), the more bandwidth is required to fully transmit that data. UHF air-to-ground voice, for instance, is a low bandwidth item.

**bit errors:** The fraction of received digital bits that are errors in a transmission. The lower the bit error, the better the quality of the transmission. Bit errors, or more precisely, bit error rate, is usually expressed in exponential notation such as $1x10^{-6}$ (one bit error in a million).

**bit rate:** The rate that digital bits of data are transmitted in a digital stream. The higher the bit rate, the faster the transmission. Bit rates can vary from kilobits per seconds (thousands of bits per second) to gigabits per second (billions of bits per second).

**boresight:** The focal axis of a directional antenna.

**Capcom:** By rule, the designated voice contact between Mission Control and the astronauts. By tradition, the Capcom is always another astronaut. Others may also, on occasion, speak with the astronauts but are never referred to as "Capcom."

**carrier frequency:** The selected frequency used to transport radio signals.

**collimation tower:** A tower, usually located a few kilometers from the main ground station antenna, equipped with a radio frequency emitter used as an aim-point to checkout and calibrate the automatic tracking capability of the antenna.

**data rate:** The rate of downlink or uplink between a spacecraft and its ground station. Usually measured in bits per second.

**demodulate:** The removal of the modulation on a carrier frequency using a series of electronic filters so as to isolate the data from its carrier.

**digital:** A transmitted radio frequency signal or scheme comprised only of ON and OFF pulses (0's and 1's).

**eccentricity:** The amount of oblateness in a spacecraft's orbit. A perfectly circular orbit has zero eccentricity while a high apogee orbit that swings around, for example, to the back side of the Moon is of high eccentricity.

**electromagnetic waves:** Electromagnetic (EM) waves or radiation is a self propagating wave in space with electric and magnetic components. Such waves carry energy and momentum. All energetic media such as heat, light and radio transmissions are part of the electromagnetic spectrum.

**elevation:** The angle above the horizon. An antenna pointed perfectly at the horizon has zero elevation. Ninety-degrees is directly overhead.

**equatorial orbit:** An orbit which has no inclination or tilt with respect to the Equator. Thus a spacecraft in true equatorial orbit will always revolve over the Equator.

**field of view:** The amount of sky that can be viewed at any one time. For an optical device, it is what can be seen at any moment with a given aperture. For an electronic device like an antenna, it is also the region where a radio frequency link can be reasonably detected.

**Figure of Merit:** Commonly referred to as G/T, or "G-over-T", a ground station's Figure of Merit is a fundamental quantitative measure of its overall capability to enhance the received signal with respect to noise. The higher the G/T, the more sensitive the ground station is.

**g:** A measure of the equivalent gravitational acceleration experienced by an object or a person. A person standing at mean sea level experiences 1 g. Apollo astronauts returning from the Moon experienced around 8 g during reentry.

**gain:** An increase in electromagnetic signal strength due to any of several sources, resulting in the output signal being measurably stronger than the input signal. Types of gain include amplifier gain due to active electronic components such as High Power Amplifiers or Low Noise Amplifiers, and antenna gain due to antenna features such as large dish aperture and parabolic shape.

**geode:** The mathematical, gravitational model of Earth characterizing its local variations in shape, size, and mass concentrations, used by computers to calculate the orbit and trajectory of a spacecraft.

**geosynchronous orbit:** A zero-inclination, circular orbit at an altitude of 35,900 kilometers (22,300 miles) above the Equator. In such an orbit, a spacecraft's rate of revolution round Earth is the same as the rotation rate for a point on the Equator. The craft would thus appear to be hanging stationary over a given location. The term is often used synonymously with 'geostationary orbit.'

**Go/No-Go:** A decision point during a mission when Mission Control has to determine whether to proceed or abort.

**Ground Network:** A network of NASA ground stations organized under the Science Mission Directorate of Headquarters. These consist of stations in Alaska, Antarctica, Florida, Norway, and Virginia. The Ground Network (GN) also includes support from the Network Integration Center located at the Goddard Space Flight Center and the GN scheduling and VHF systems at White Sands, New Mexico.

**ground station:** A location on the ground comprising of electrical, mechanical, and/or optical subsystems used for communicating with and retrieving data from space assets.

**ground track:** The path that a spacecraft traces on the ground.

**GSTDN:** The original ground elements of the remaining Spaceflight Tracking and Data Network (STDN) as the Tracking and Data Relay Satellite System (TDRSS) was being brought operational in the 1980s.

**high inclination orbit:** An orbit whose plane is highly inclined with respect to the Equator. Such orbits have ground tracks that enable them to pass over or observe a greater amount of Earth's surface than low inclination orbits.

**high power amplifier:** An electronic device usually located at the base or the back of an antenna which greatly amplifies its transmitting signal strength for establishing command uplink with a spacecraft.

**housekeeping telemetry:** Data from a spacecraft used only for assessing the performance, health, and status of the spacecraft itself. Typical data include voltages, temperatures, propellant tank pressures, etc.

**Lagrange Points:** The Lagrangian points are five positions in the Earth-Moon-Sun interplanetary system where a small object affected only by gravity can theoretically appear stationary relative to two larger objects (such as a satellite with respect to Earth and the Moon). Such an object would appear to be in a "fixed" position (or on a relatively small "Figure 8" orbit) in space rather than in a traditional orbit in which its position changes continuously.

**launch azimuth:** The direction that a rocket is launched in, usually measured in degrees relative to due North. For example, a rocket launched due East from Cape Canaveral has a launch azimuth of 90°.

**"Lights-out" operations:** A trend in ground station operations in which a station operates autonomously in an unattended fashion or with minimal staffing. Lights-out can take many forms such as nine to five workday operations with automated service at all other times. It can also be a centralized, fully staffed operations center continuously monitoring a suite of multiple, unattended remote ground stations.

**line of sight:** The straight line between a transmitting antenna and a receiving antenna. The two can only communicate when a line of sight has been established.

**multipath:** The propagation phenomenon that results when radio signals reach the receiving antenna by two or more paths. Causes of multipath include ionospheric reflection and refraction, and reflection from mountains, buildings, ocean, and the ground. The effects of multipath include constructive and destructive interference, errors, and phase shifting of the signal. A common multipath phenomenon is ghosting of television images, for example.

**"Picket Line":** The seven original primary Minitrack stations located approximately along the 75th West meridian, spanning North and South America. The picket line had a better than 90 percent chance of capturing every pass of a satellite in low inclination, low-Earth orbit.

**polar orbit:** An orbit whose plane is inclined 90° to the Equator. Thus, as its name implies, a satellite in polar orbit travels over, or near, both the North and the South pole. A great advantage of a satellite in such an orbit is its ability to observe the entire surface of the globe over time.

**max-q:** The occurrence of maximum aerodynamic pressure (q) during a rocket's ascent or a spacecraft's descent through the atmosphere. Knowing when max-q occurs is important as it factors into the structural stress experienced by the vehicle.

**parabolic:** Surface shape of an object, like an antenna reflector or the objective mirror in a telescope, based on the mathematical curve $Y=X^2$ which focuses incoming rays to a single point.

**radio frequency:** The number of oscillation cycles per unit time that an electromagnetic wave propagates through space at. Frequency is measured in hertz (Hz).

**radio interferometry:** The use of separate antennas to receive a radio signal at slightly different times so as to determine phase differences in the radio waves. These differences can then be used to calculate position solutions for an object. Radio interferometry had the advantage of yielding highly accurate tracking angles and could be used under virtually any atmospheric condition.

**Schmidt camera:** A telescopic, astronomical camera designed to provide wide fields of view, typically used as a survey instrument in which a large amount of sky must be covered.

**solar wind:** A stream of charged particles (plasma) which are ejected from the upper atmosphere of a star. It consists mostly of high-energy electrons and protons that are able to escape the star's gravity. Many phenomena can be explained by the solar wind, including: geomagnetic storms that knock out power grids on Earth; auroras, and why the tail of a comet always points away from the Sun.

**sounding rocket:** A small, usually unguided rocket launched into the upper atmosphere for conducting experiments and scientific research.

**Space Network:** NASA's constellation of geosynchronous Tracking and Data Relay Satellites and their associated ground segment. The ground segment consists of the White Sands Complex, the Bilateral Ranging Transponder System, the Merritt Island Launch Area TDRSS Relay, and the Data Services Management Center. The Space Network (SN) is run by the Goddard Space Flight Center under management of the Space Operations Mission Directorate at Headquarters.

**state vectors:** The set of position and velocity measurements of a traveling object as a function of time, particularly those of a spacecraft in orbit or a rocket on a ballistic trajectory.

**telemetry:** Electronic data measurements downlinked or transmitted from a spacecraft to the ground.

**teletype:** A now largely obsolete electro-mechanical typewriter which was used to communicate typed messages from point to point through a simple electrical communications channel, often just a pair of wires. Later versions used a screen instead of a printer.

**tracking:** Collection of spacecraft position and velocity measurements so that its orbit or trajectory can be determined.

**wavelength:** The distance traveled by an electromagnetic wave during one cycle of oscillation. Radio frequency wavelengths ranges from millimeters to meters.

**Wing Station:** A support or backup ground station located near the designated primary station on a given mission. Deep Space Network sites were often used to support nearby Manned Space Flight Network sites in this manner.

**yagi antenna:** An antenna consisting of an array of linear elements, such as a common rooftop television antenna. The antenna achieves a distinct response bandwidth determined by the length, diameter, and spacing of all the individual elements. Its overall gain is proportional to its length, rather than simply the number of elements. Yagis can range in size from small (like TV antennas) to very large (dozens of feet long with multiple elements).